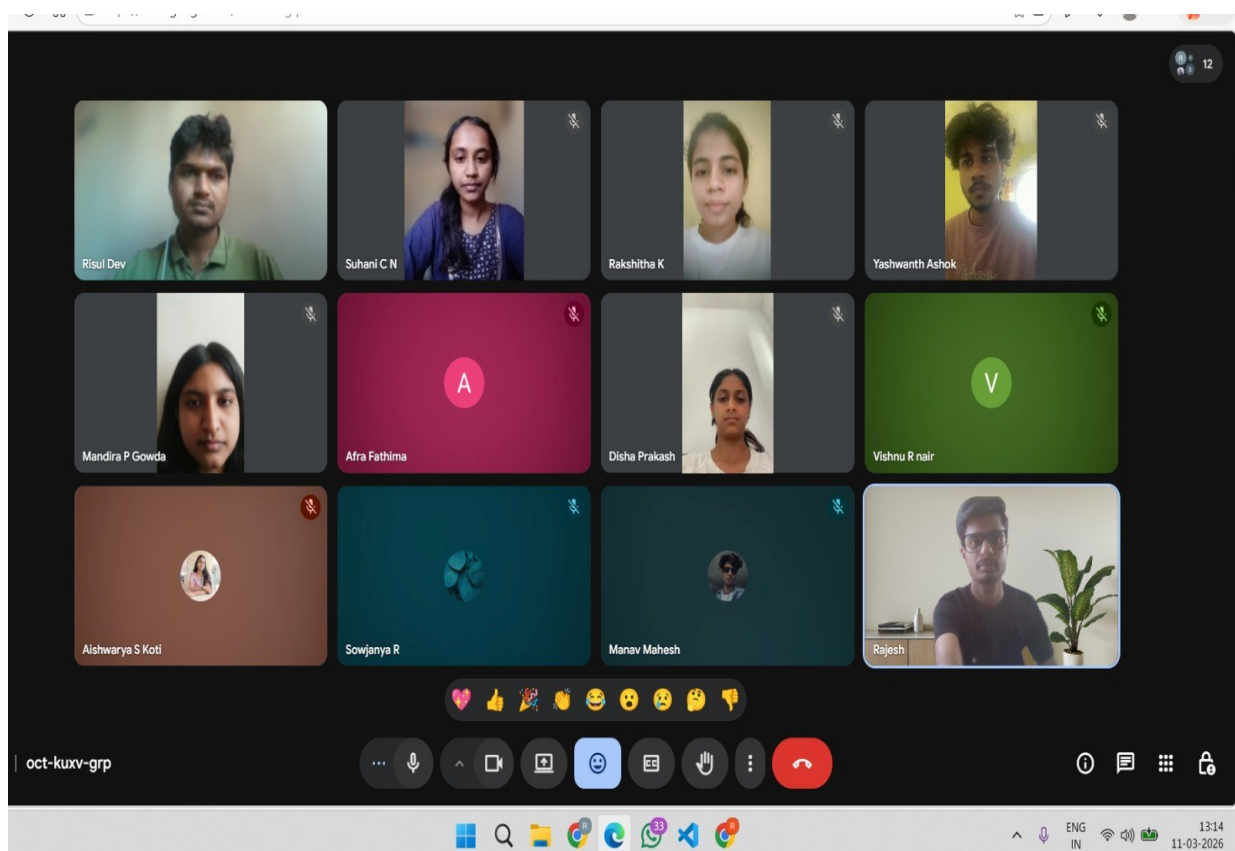


Mini Project: Cohort 21: Forest Fire Ignition Risk

Date: 11/03/2026

On Day 5 of the project, the team continued progressing with the **model evaluation and comparison phase**, which is a critical step in determining the best-performing model for predicting forest fire ignition risk. Following the successful implementation of the Decision Tree model on Day 3, the team proceeded to build and evaluate the **Random Forest model** as planned. The **Algerian Forest Fire Dataset** continued to serve as the foundation for model training and testing.



During the session, the team focused on running both the **Decision Tree** and **Random Forest** classifiers on the prepared dataset and comparing their performance metrics. The evaluation team analyzed key metrics including accuracy score, confusion matrix, and the classification report for both models. This comparative analysis was important to identify which model generalizes better on unseen data and provides more reliable predictions.

The Random Forest model, which is an ensemble learning technique, builds multiple decision trees during training and combines their outputs to produce a more accurate and stable prediction. Unlike a single Decision Tree, Random Forest reduces the risk of overfitting by

averaging the results of several trees trained on different subsets of the data. The team observed that the Random Forest model demonstrated improved accuracy compared to the standalone Decision Tree, making it a strong candidate as the final model.

Key Discussion Points

Model Comparison: The team compared the accuracy scores of both model's side by side. The Random Forest model showed higher overall accuracy and better performance across all fire risk categories present in the dataset.

Hyperparameter Tuning: The team discussed potential hyperparameter tuning options for the Random Forest model, such as adjusting the number of estimators (trees), maximum depth, and minimum samples split, to further improve model performance.

Feature Importance: The Random Forest model also provided insights into feature importance, allowing the team to identify which meteorological and environmental factors contribute most significantly to predicting forest fire ignition risk in the Algerian dataset.

Next Steps

Based on the evaluation results from Day 5, the team has shortlisted the **Random Forest model** as the preferred final model for the project. The next steps involve finalizing the model, performing any remaining optimizations, and preparing the project report and presentation. The evaluation team will document the complete performance metrics and findings to support the final submission.